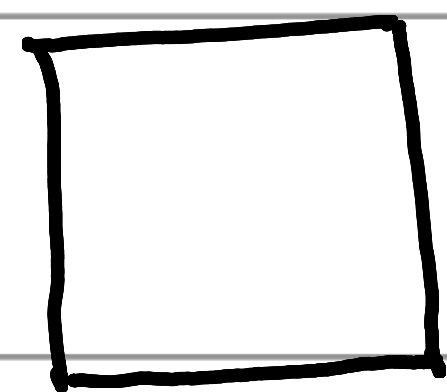


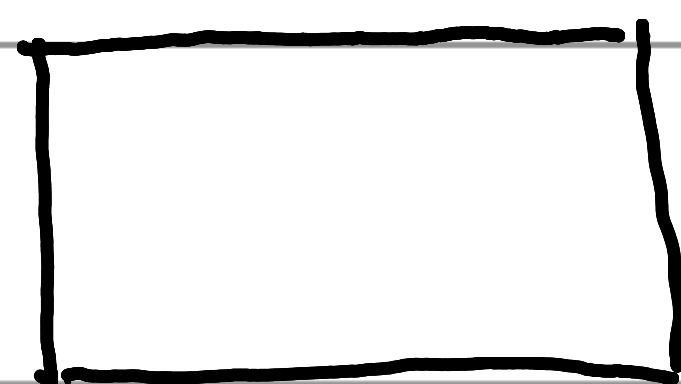
23 Sep' 25 // 28 Sep' 26

## Geometry — Shapes



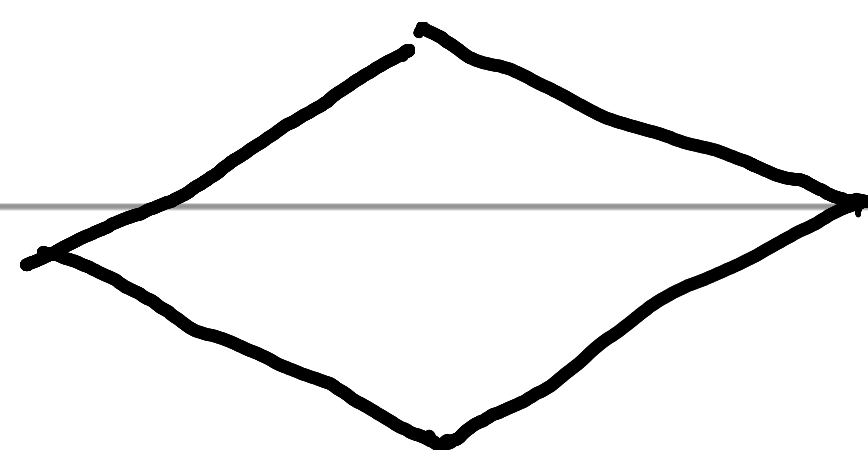
Square

All four sides are equal  
The sides should be able to  
be aligned horizontally  
and vertically



Rectangle

Has four sides; Horizontal  
and vertical; Opposite  
sides are equal

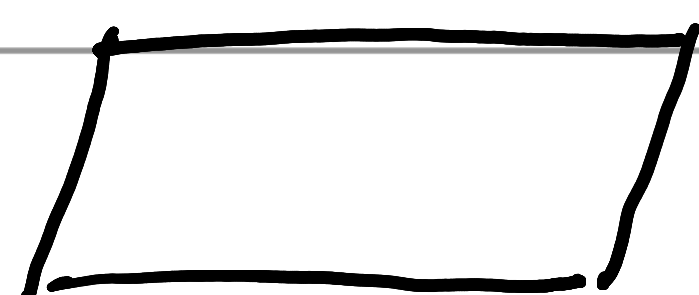


Rhombus

All four sides are equal;  
Can be aligned in any  
direction unlike square

Every square is a rhombus but

every rhombus is not a square



Parallelogram

4 sides; Opposite sides are  
equal; Can be aligned  
in any direction.

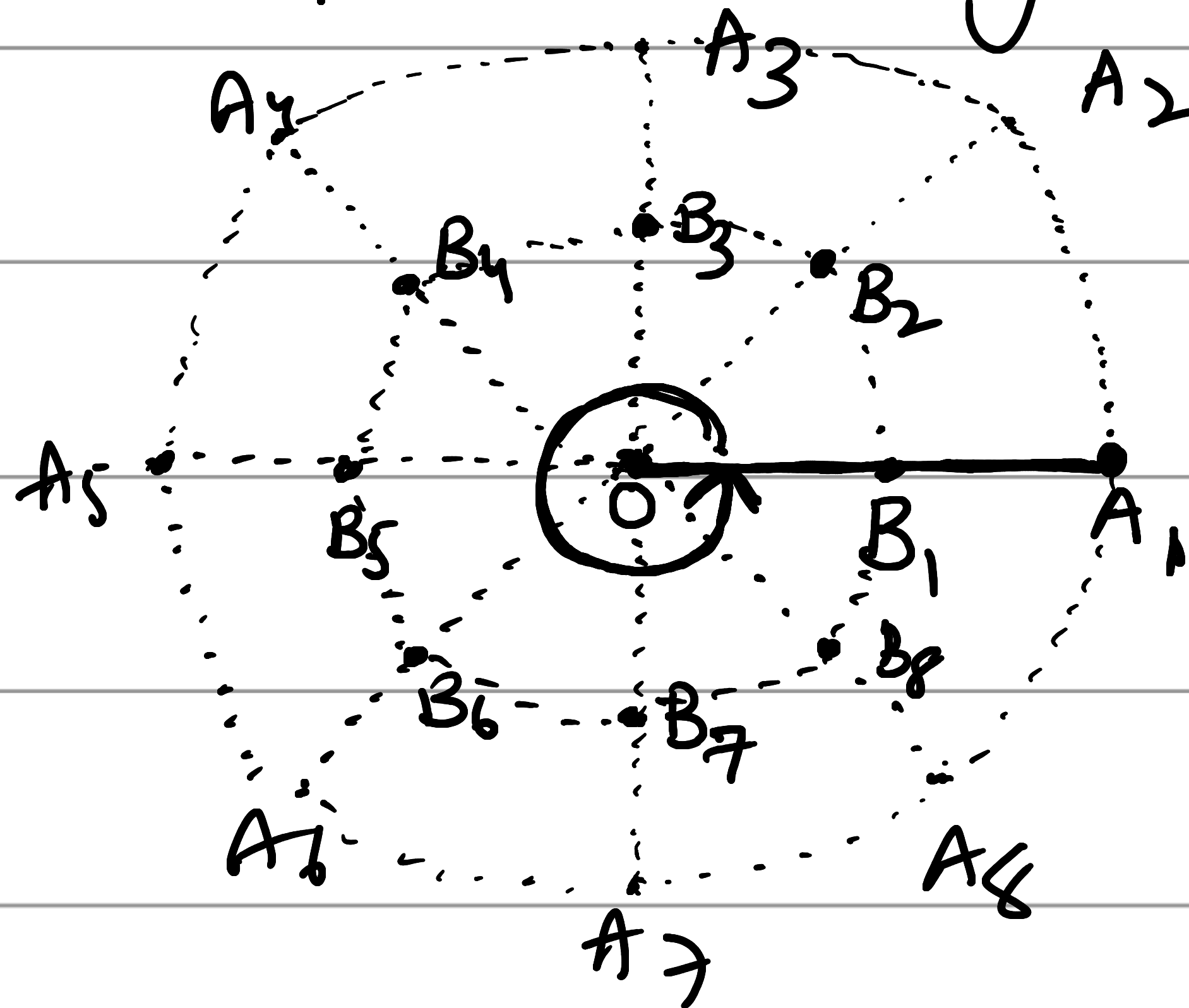
Every rectangle is a parallelogram but  
not vice versa.



→ Triangle

- 3 sides
- and
- 3 angles

What is angle?



(Think 'O' is one end  
of a stick fixed to the ground; 'A'  
is the free end of the stick).

If we rotate  $OA_1$  around  
O, we get a circle.  
Angle tells us how  
much we have rotated.

Full rotation brings it back  
to original position ( $OA_1$ ).

If  $B_1$  is the mid point of O and  $A_1$ ,  
then  $B_1$  will also make a circle when  $A_1$  makes  
a circle around O.

Half rotation moves it to the opposite side ( $OA_5$ ).

$\frac{1}{4}$ th rotation takes it from horizontal to vertical ( $OA_3$ )

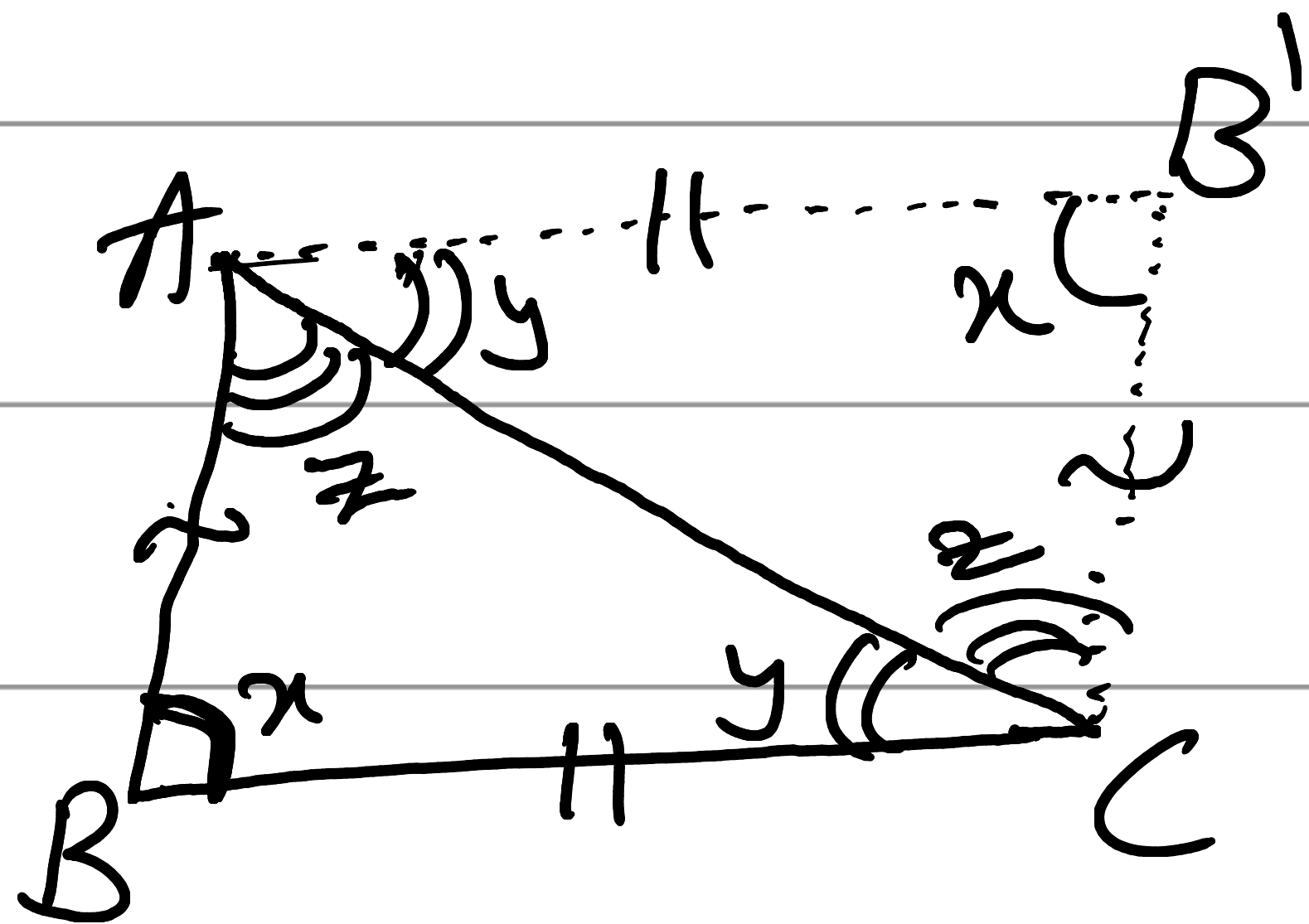
So, in squares and rectangles all the angles are  $\frac{1}{4}$ th of a full rotation.

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→ See AI notes on Geometry - Introduction

Theorem: Sum of angles in a triangle is  $180^\circ$

Proof: Let  $\triangle ABC$  be any triangle.



Let  $B'$  be another point in the plane

Such that  $\overline{AB'} = \overline{BC}$  &  $\overline{AB} = \overline{B'C}$

So  $ABCB'$  is parallelogram

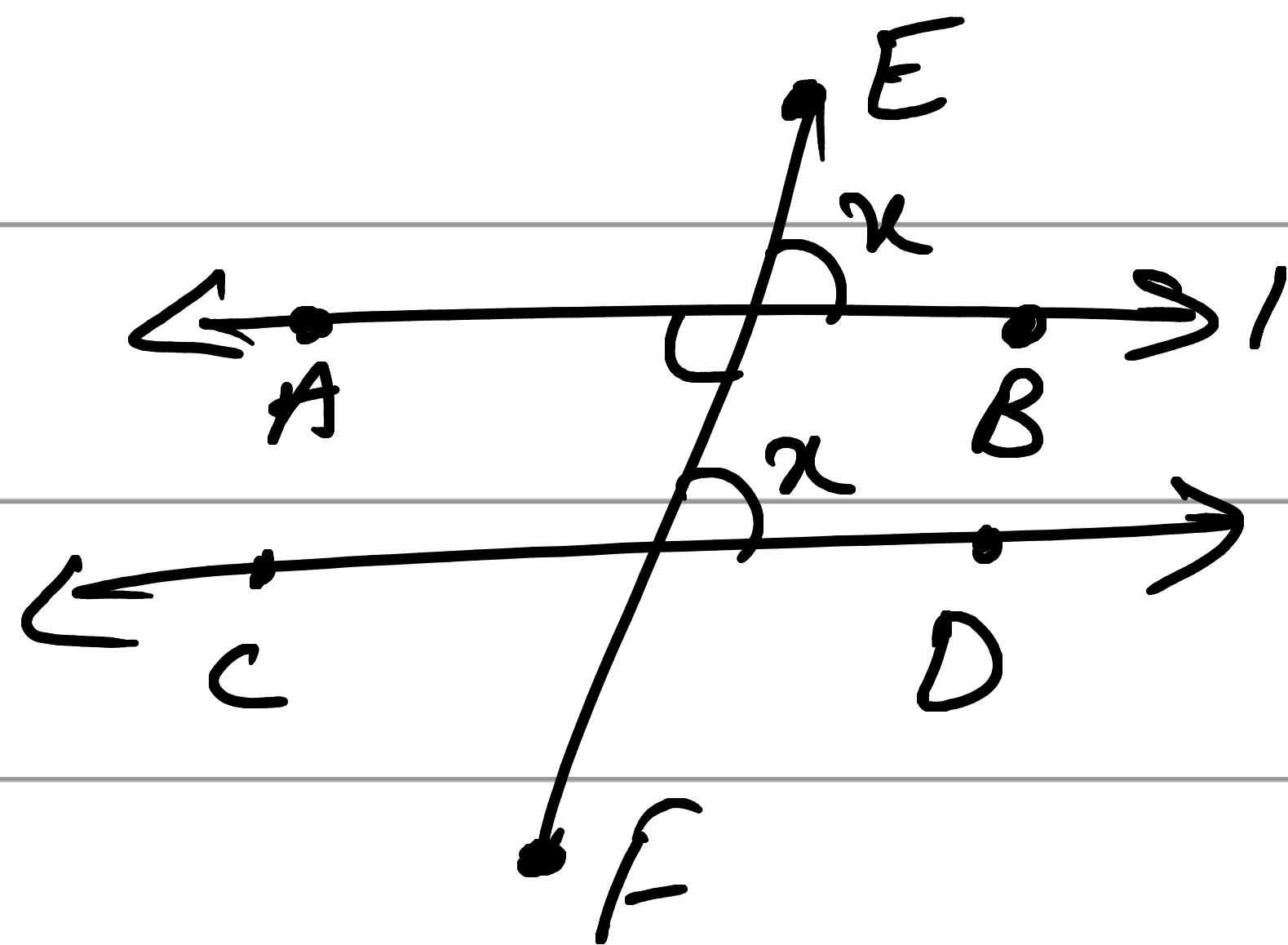
Sum of angles in a parallelogram =  $360^\circ \implies$  Why?  
 $\angle B + \angle C + \angle B' + \angle A = x + (y+z) + x + (y+z) = 360^\circ$

$$\implies 2(x+y+z) = 360^\circ$$

$\implies x+y+z = 180^\circ \implies$  Sum of angles of the triangle =  $180^\circ$

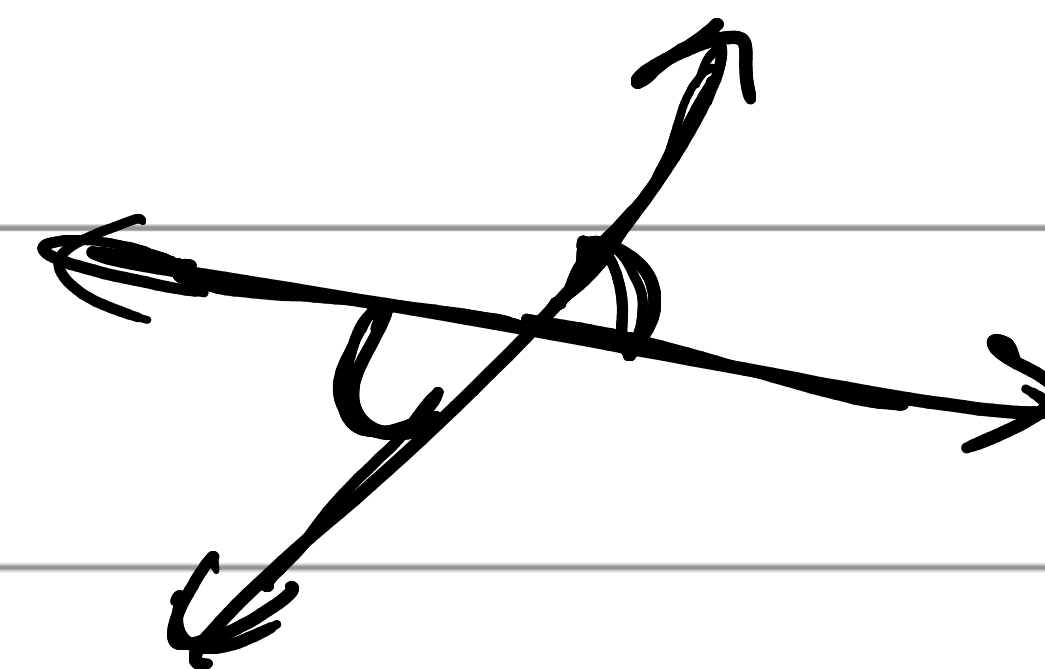
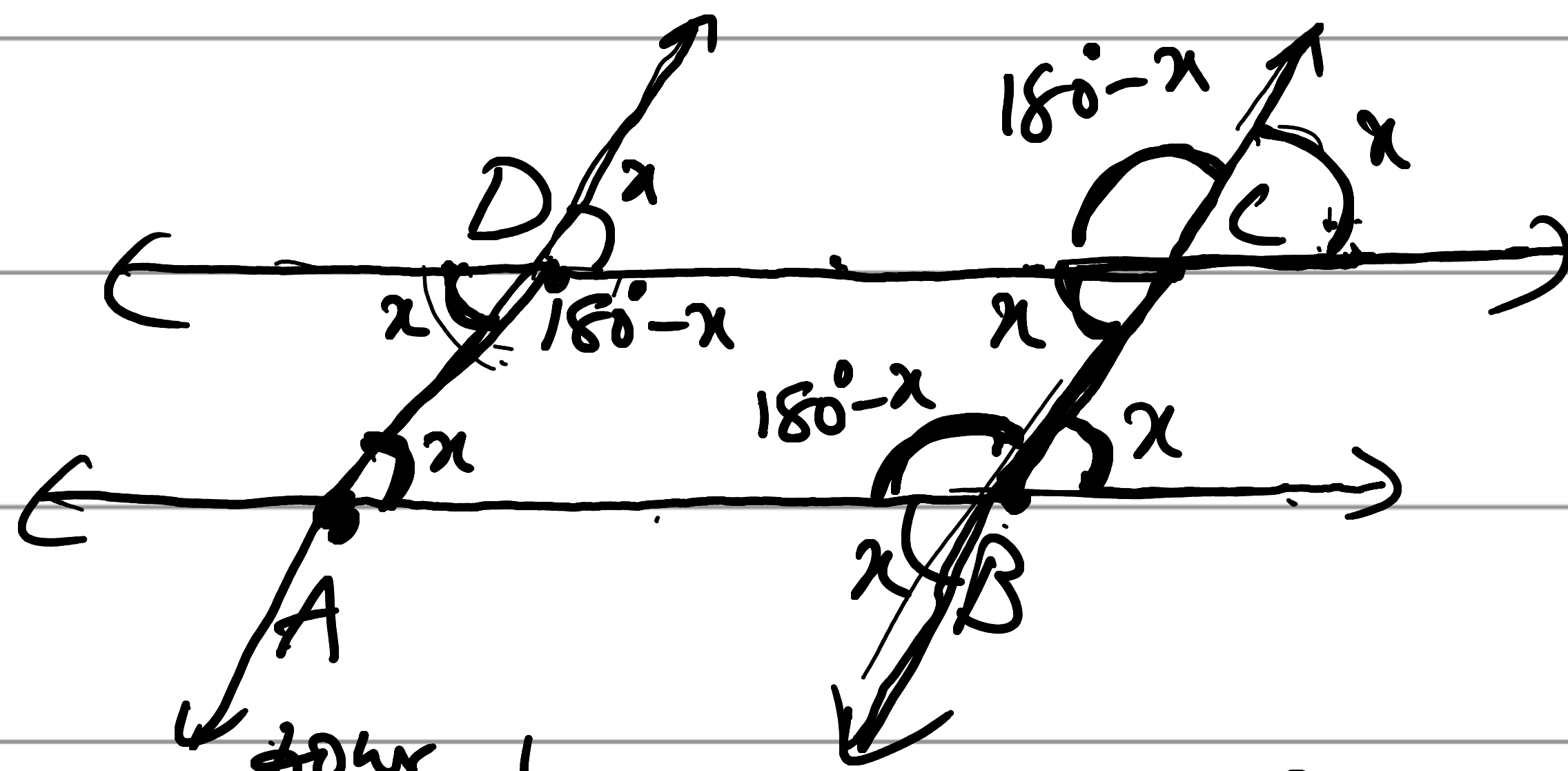
Theorem:- Sum of the angles in a parallelogram is  $360^\circ$ .

Proof



A line segment  $\overline{EF}$  intersects two parallel lines  $\overleftrightarrow{AB}$  and  $\overleftrightarrow{CD}$  and makes the same angle ' $x$ ' at both the parallel lines

Let take a parallelogram ABCD



$$\begin{aligned} \text{Sum of the angles in } ABCD &= x + (180^\circ - x) + x + (180^\circ - x) \\ \angle A + \angle B + \angle C + \angle D &= 360^\circ \end{aligned}$$

# Axioms:

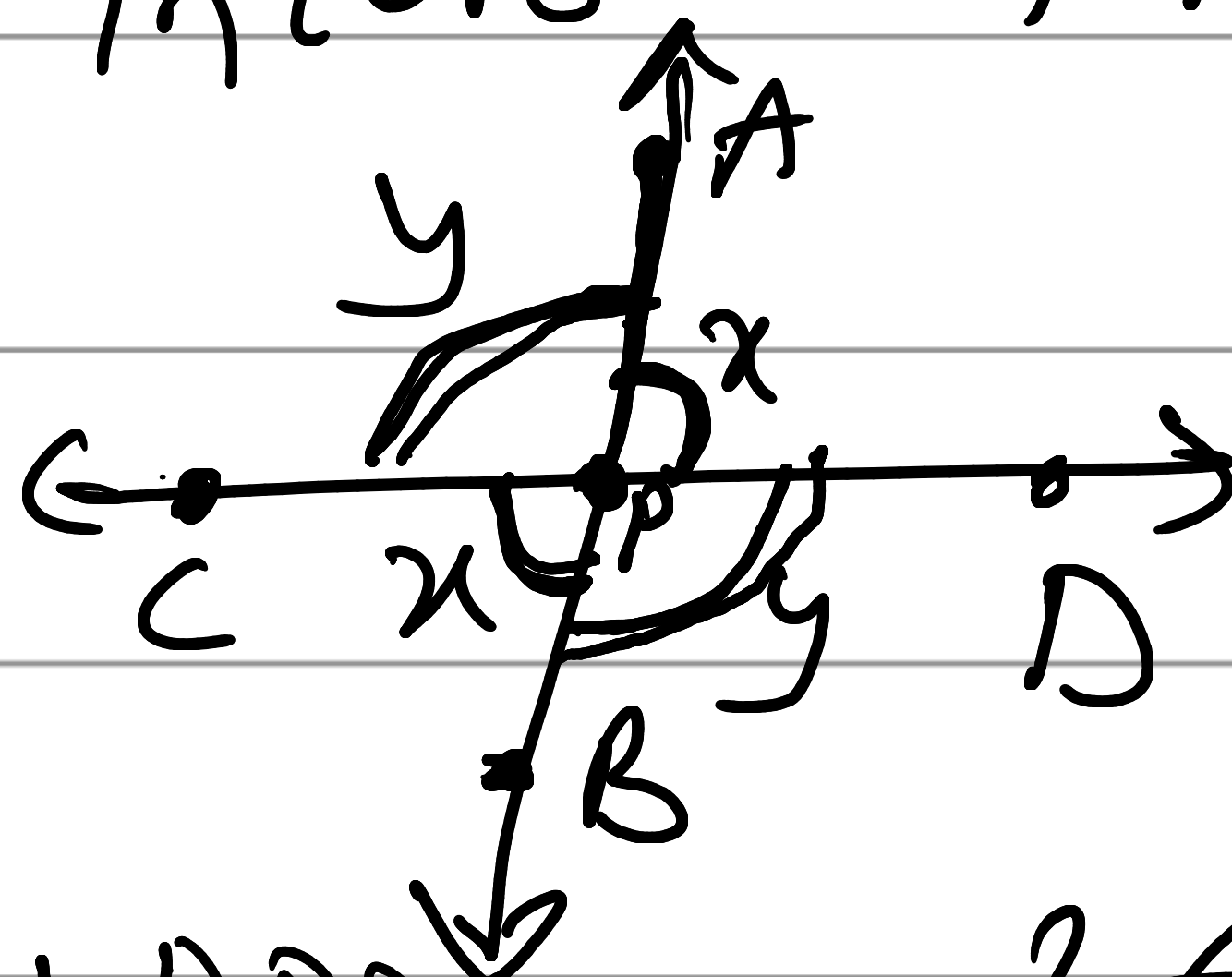
(1)  $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$   
(parallel to)



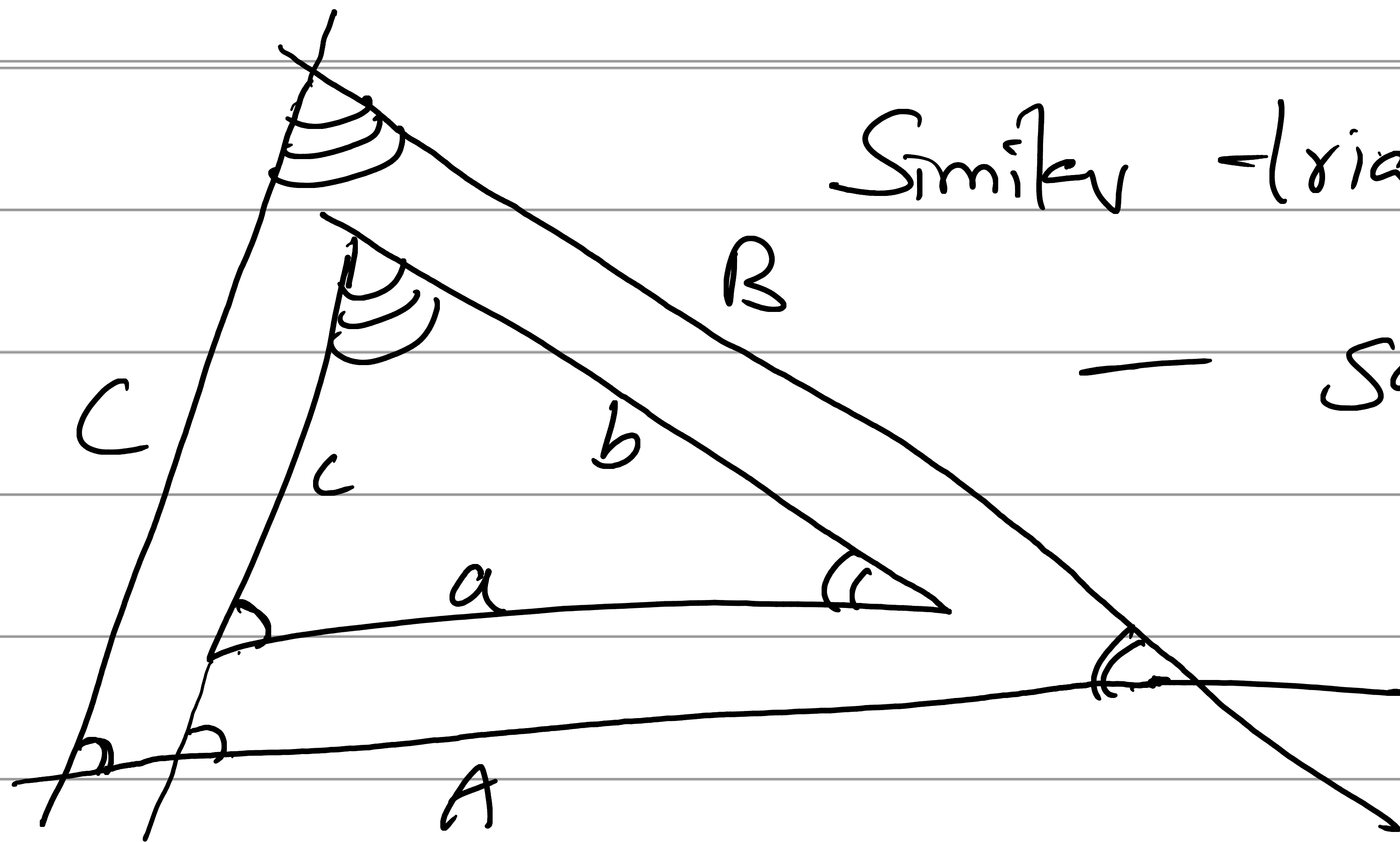
$\overleftrightarrow{EF}$  intersects  $\overleftrightarrow{AB} \cap \overleftrightarrow{CD}$  at  $P \cap P'$ , respectively.

Then  $\angle EPB = \angle EP'D$

(2) If  $\overleftrightarrow{AB}$  intersects  $\overleftrightarrow{CD}$ , then at their point of intersection 'P', opposite angles are equal.



$x + y = \angle APD + \angle DPB = \angle APB = \text{half rotation at 'P'} = 180^\circ$ .

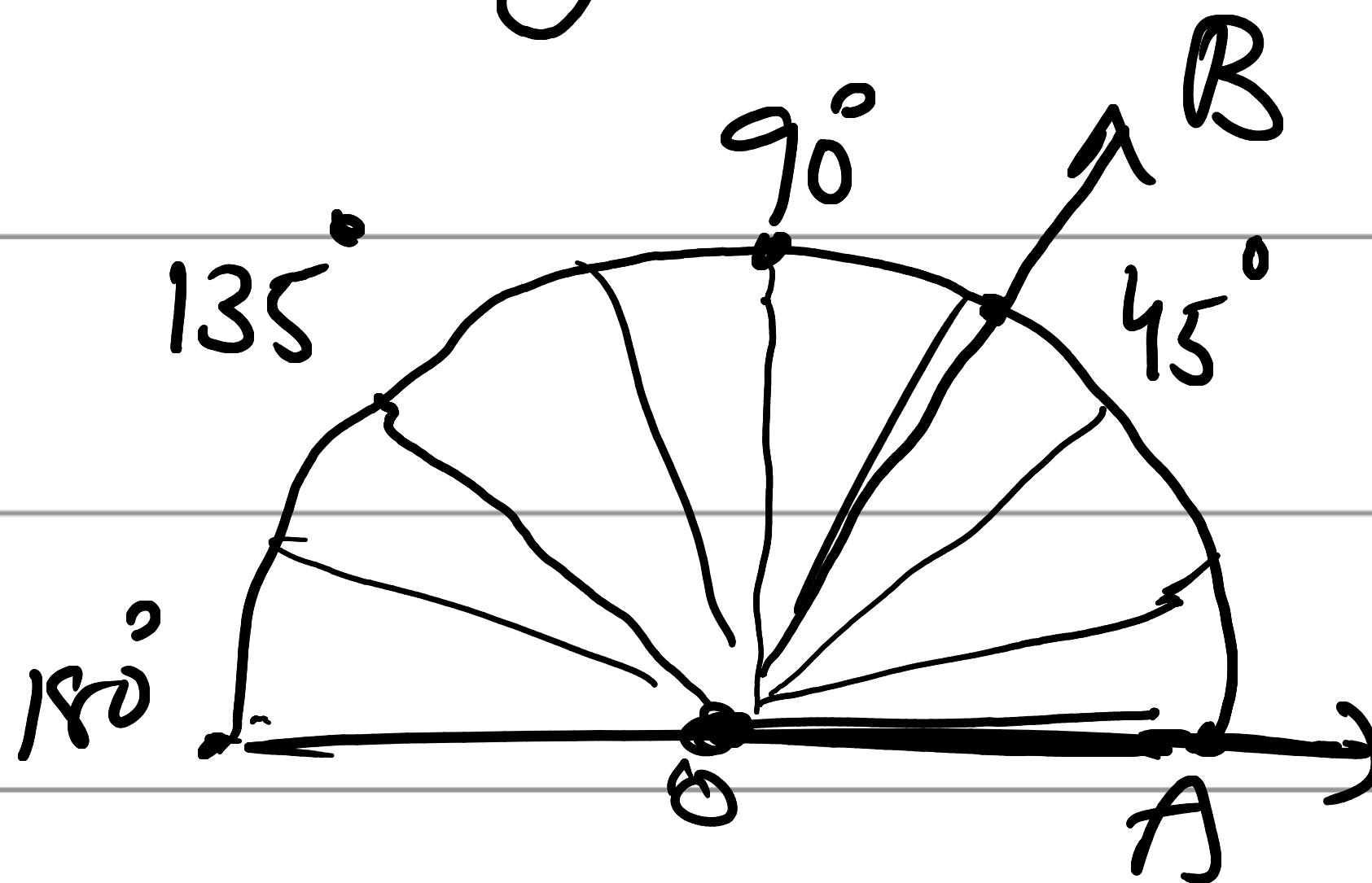
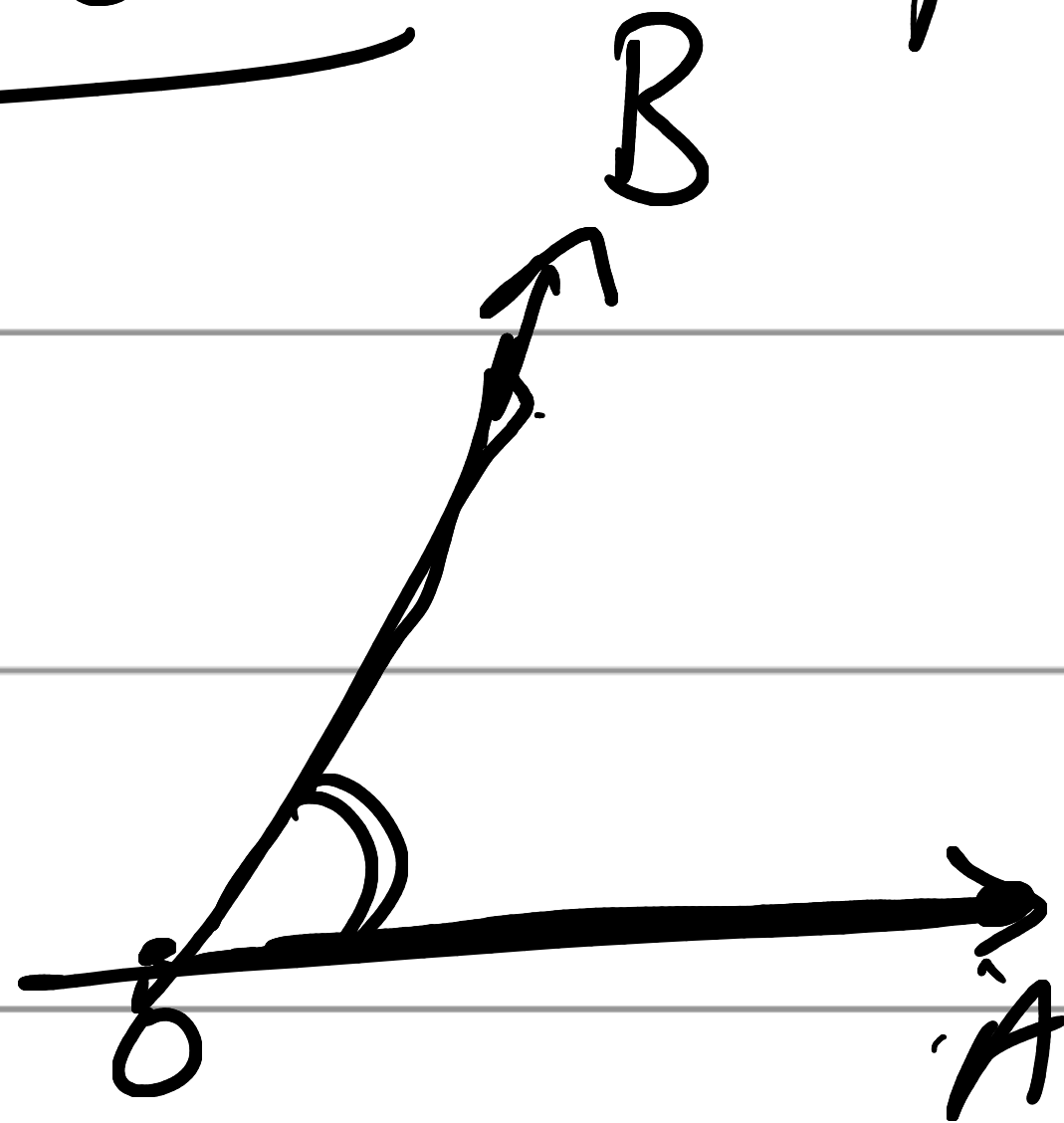


Similar triangles

— Same angles  
but different  
lengths  
of sides.  
(proportion)

$$\frac{a}{A} = \frac{b}{B} = \frac{c}{C}$$

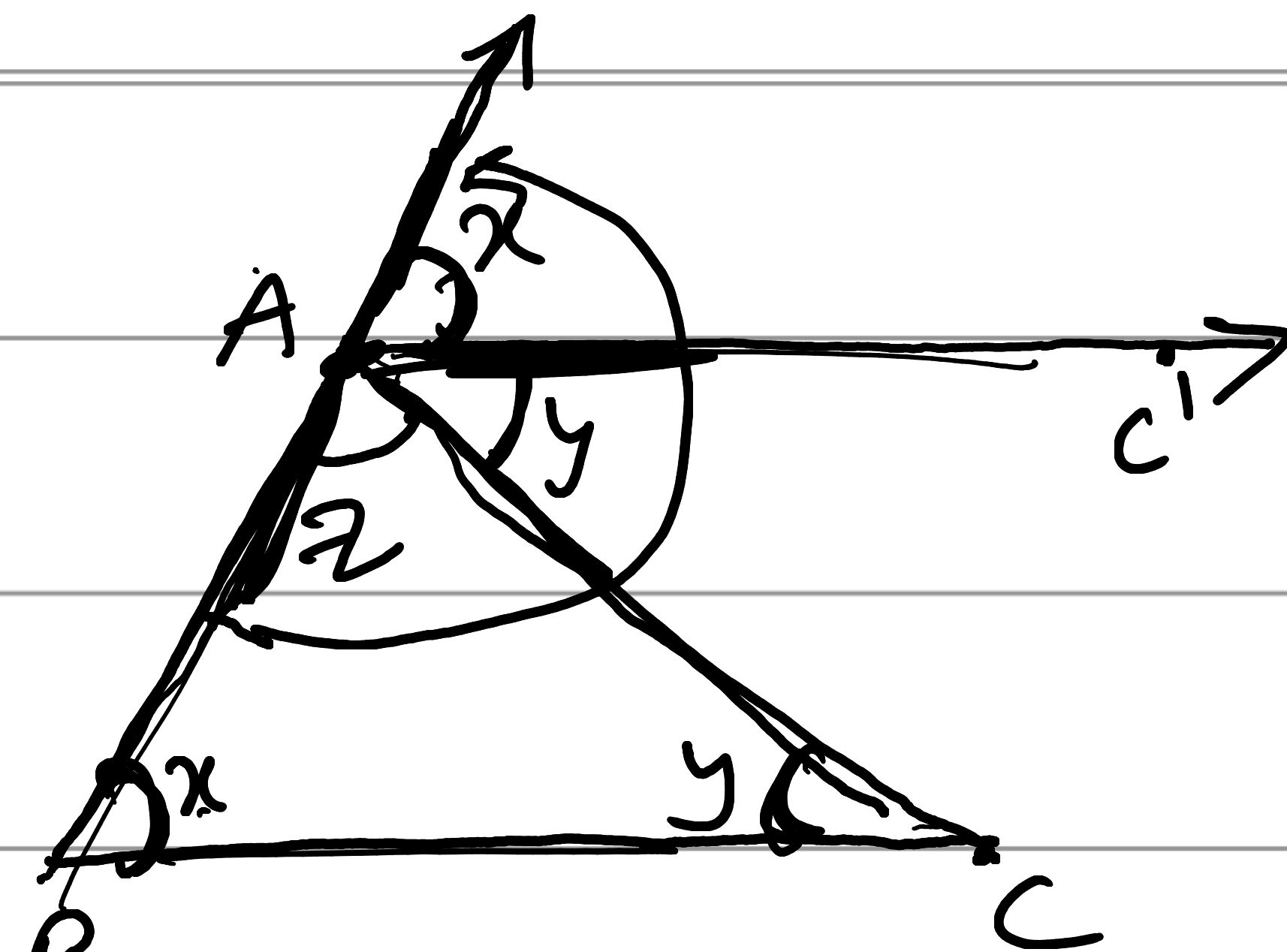
Protractor: Helps measure angles.



→ Sum of angles

in a

triangle:

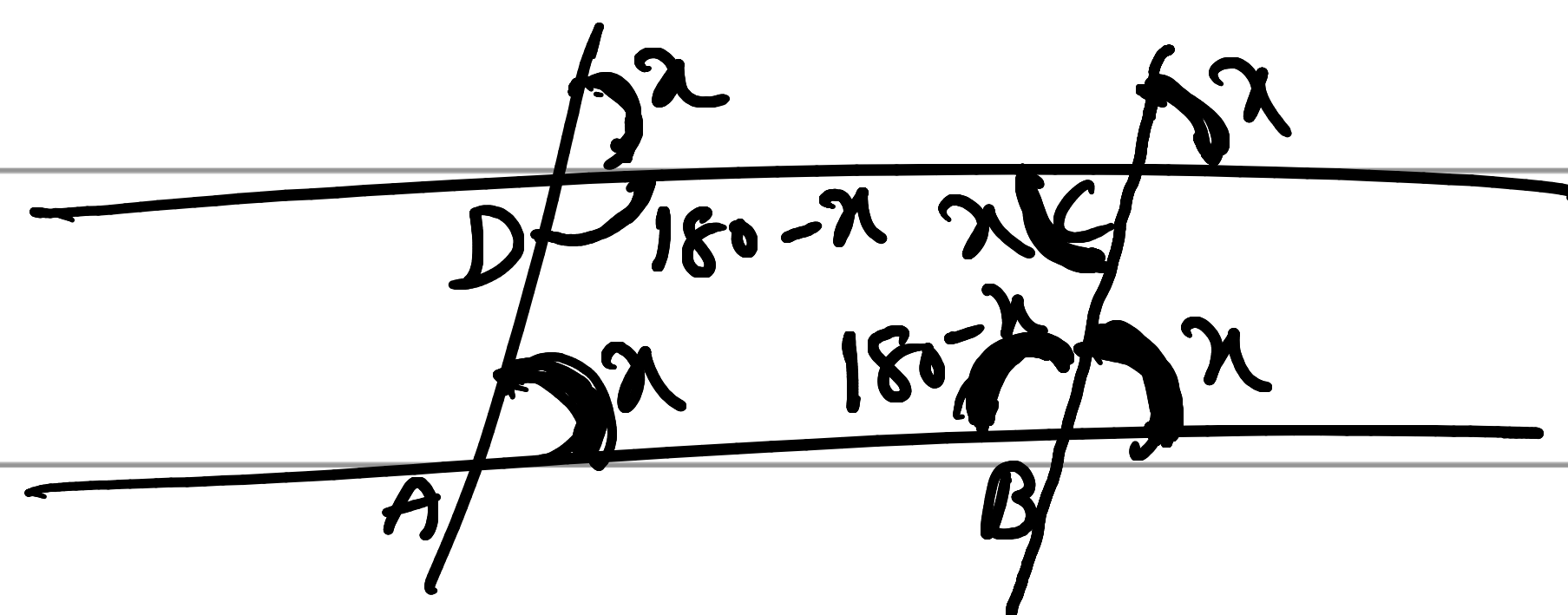


$$\vec{AC'} \parallel \vec{BC}$$

$$x + y + z = 180^\circ$$

(see AI description)

→ Sum of angles in a //gram



$$\begin{aligned} \text{Sum} &= \angle A + \angle B + \angle C + \angle D = x + (180^\circ - x) + x + (180^\circ - x) \\ &= 360^\circ \end{aligned}$$

HW: Read AI notes, class notes, solve problems, homework exercises.

HW: Buy a Geometry box

Todos for next class:

(1) Areas

(2) Circle —  $\pi$

(3) Sets — see AI notes